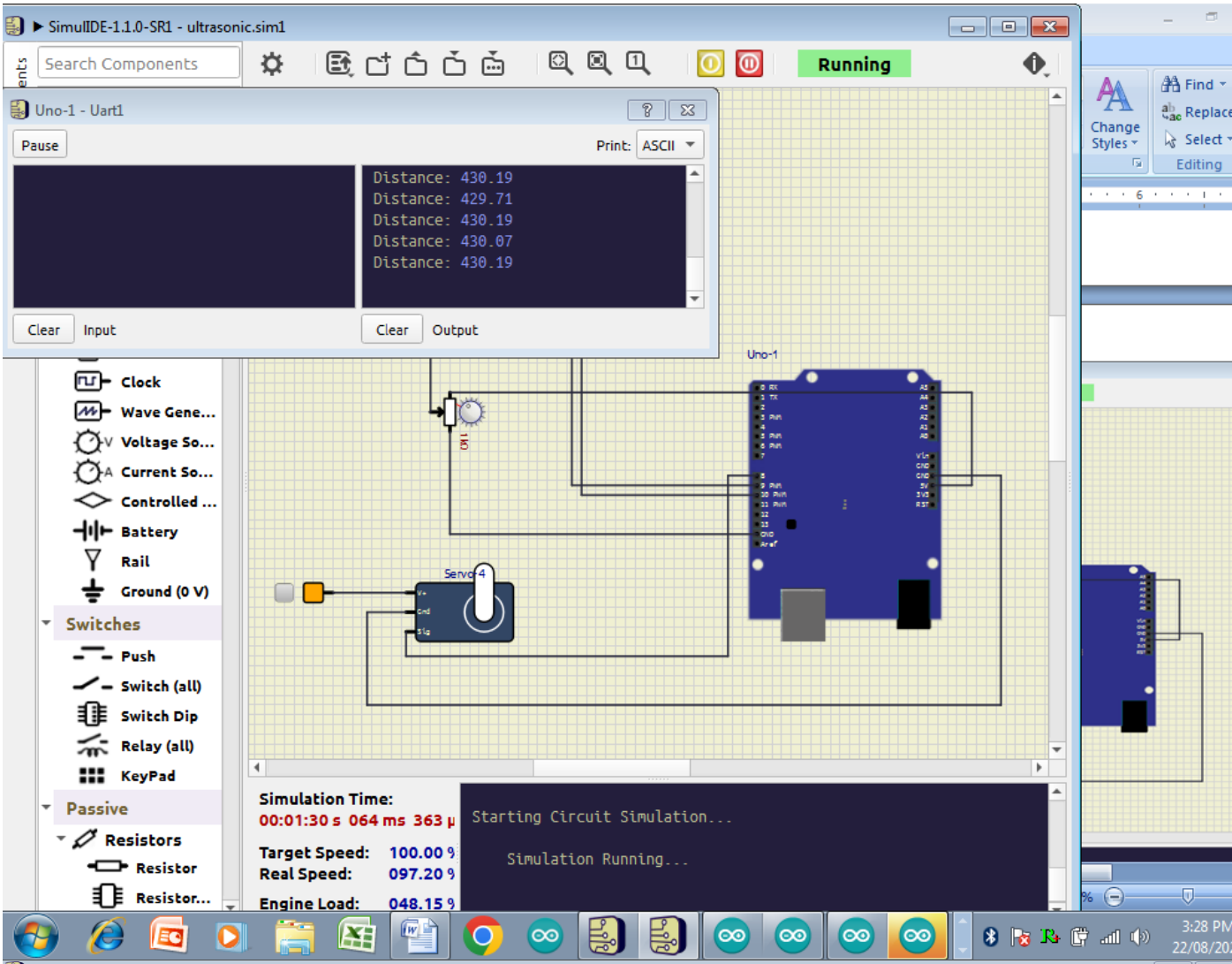
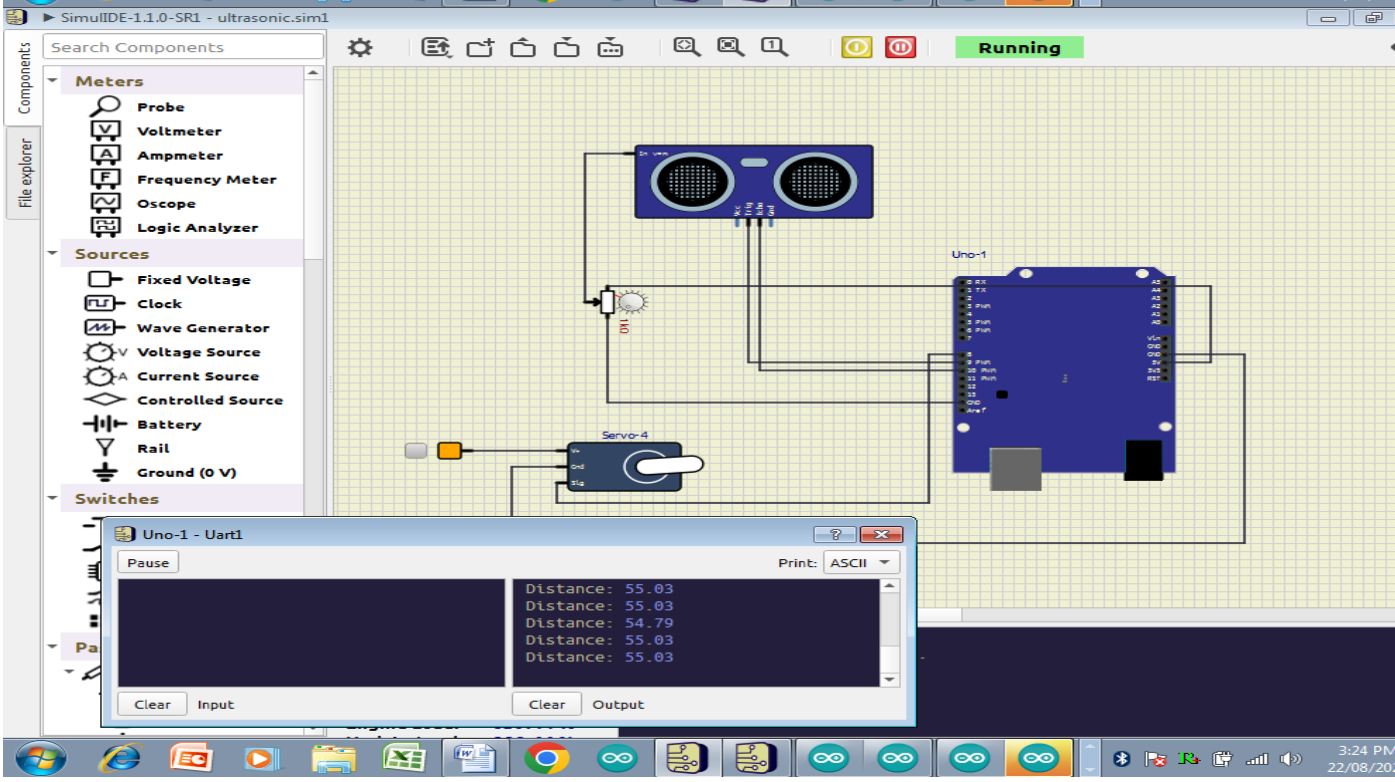
```
#include <Servo.h>

Servo myservo;
int pos = 0;
const int trigPin = 9;
const int echoPin = 10;
int b1=4;
float duration, distance;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  Serial.begin(9600);
  myservo.attach(8);
}

void loop() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  duration = pulseIn(echoPin, HIGH);
  distance = (duration*.0343)/2;
  Serial.print("Distance: ");
  Serial.println(distance);
  delay(100);
    if(distance>0 && distance<250)
     {
      myservo.write(90);
      delay(3);
       myservo.write(0);
      delay(3);
     }
        else
     {
       myservo.write(90);
      delay(3);
}
}
```


SimulIDE-1.1.0-SR1 - ultrasonic.sim1
Search Components
Running
Uno-1 - Uart1
Pause
Print: ASCII
Distance: 430.19
Distance: 429.71
Distance: 430.19
Distance: 430.07
Distance: 430.19
Clear
Input
Clear
Output
Clock
Wave Gene...
Voltage So...
Current So...
Controlled ...
Battery
Rail
Ground (0 V)
Switches
Push
Switch (all)
Switch Dip
Relay (all)
KeyPad
Passive
Resistors
Resistor
Resistor...
Uno-1
Servo-4
Simulation Time:
00:01:30 s 064 ms 363 µ
Target Speed: 100.00 %
Real Speed: 097.20 %
Engine Load: 048.15 %
Starting Circuit Simulation...
Simulation Running...
Find
Replace
Select
Change Styles
Editing


SimulIDE-1.1.0-SR1 - ultrasonic.sim1
Search Components
Running
Components
File explorer
Meters
Probe
Voltmeter
Ampmeter
Frequency Meter
Oscope
Logic Analyzer
Sources
Fixed Voltage
Clock
Wave Generator
Voltage Source
Current Source
Controlled Source
Battery
Rail
Ground (0 V)
Switches
Uno-1
Servo-4
Uno-1 - Uart1
Pause
Print: ASCII
Distance: 55.03
Distance: 55.03
Distance: 54.79
Distance: 55.03
Distance: 55.03
Clear
Input
Clear
Output